CUSTOMER ANALYTICS

Customer Shopping Behavior Analysis

Purchase patterns, category concentration, and practical next steps.

A descriptive read of 3,900 customer records

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CATEGORY CONCENTRATION

Clothing and Accessories drive 76.6% of purchase amount

The two leading categories account for most purchase amount, making assortment and promotional planning the clearest commercial focus.



Clothing

$104,264

44.7% of total purchase amount

Accessories

$74,200

31.8% of total purchase amount

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BASELINE

The dataset shows a stable mid-market purchase pattern

Purchase amounts cluster around a $60 midpoint, with a broad but bounded range.

Records

3,900

customer records

Purchase amount

$233,081

total

Typical basket

$59.76

average · $60 median

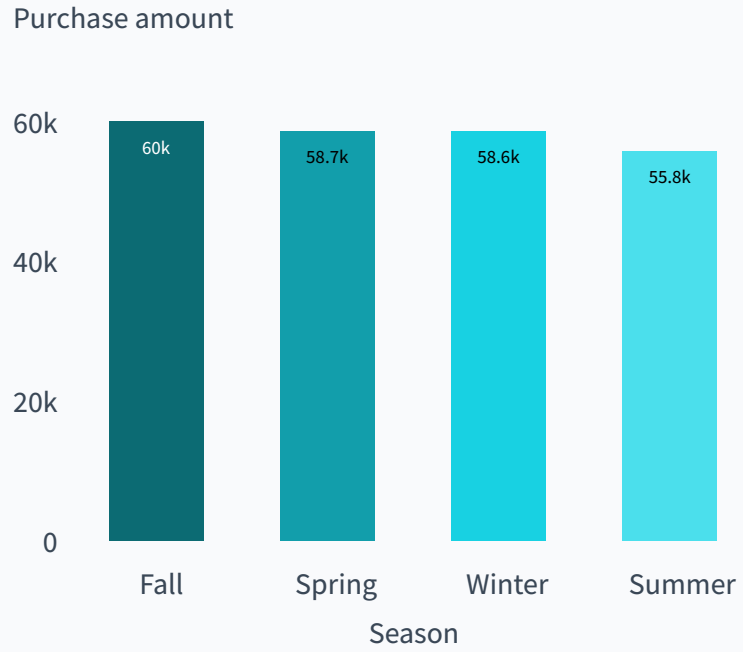
18 columns · Purchase amounts range from $20 to $100 · Average customer age: 44.1 years

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SEASONALITY

Fall leads, but the seasonal gap is modest

Fall is highest, while Summer is lowest. The spread is modest enough that seasonal campaigns should be tested rather than assumed to have a dramatic effect.



Fall

$60,018

Summer

$55,777

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SUBSCRIPTION

Subscription uptake is low, without a higher average basket

Subscription status separates record counts, but not average purchase amount.

Subscribed

1,053

27.0% of the dataset

Average purchase amount: $59.49

Not subscribed

2,847

73.0% of the dataset

Average purchase amount: $59.87

Retention benefits need separate analysis beyond average transaction amount.

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GENDER MIX

Gender revenue differences reflect record mix more than basket size

Total revenue follows the number of records. Average purchase amounts are similar.

Male records

$157,890

2,652 records

Average purchase amount: $59.54

Female records

$75,191

1,248 records

Average purchase amount: $60.25

Interpret total revenue alongside record counts and average purchase.

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DISCOUNTING

Discounts are common, but not linked to larger purchases here

Discounted records show a slightly lower average purchase amount in this dataset.

Records with discount

43.0%

of records

With discount

$59.28

average purchase amount

Without discount

$60.13

average purchase amount

This is an association, not proof that discounts reduce spend. Evaluate promotions using margin and incremental-sales measures, not revenue alone.

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ADDITIONAL CONTEXT

Shipping and reviews add useful context

Context	Value	Read-through
Shipping	$60.73	2-Day Shipping, highest average purchase amount
Shipping	$58.46	Standard Shipping, lowest average purchase amount
Review rating	3.75/5	Mean available rating
Missing ratings	37	About 0.95% of rows
Highest item rating	3.86	Gloves, interpreted cautiously

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ANALYTICAL WORKFLOW

The workflow connects Python, SQL, and Power BI

Each tool carries a different part of the analysis, from preparation to communication.

01

Python / Pandas

Inspect the CSV, standardize fields, create derived fields, and calculate summary statistics.

02

MySQL

Aggregate, filter, segment, and answer business questions with SQL.

03

Power BI

Communicate KPIs, category revenue, seasonal revenue, discount usage, and subscription status.

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NEXT STEPS

Next steps should focus on retention, margin, and repeat behavior

Prioritize Clothing and Accessories in assortment and promotional planning.

Test targeted subscription campaigns using conversion, repeat purchase frequency, retention, and customer lifetime value.

Compare seasons with product-, discount-, and customer-level results before increasing seasonal spend.

Document how missing review ratings are handled and make dashboard definitions and USD labels explicit.

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